Hafsa Sheikh

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Education

ETH Zurich Sept. 2025 – ongoing
MSc Computer Science *Zurich, Switzerland*

- **Major: Data Management Systems, Minor: Programming Languages & Software Engineering**
- **Relevant Coursework:** Big Data, Cloud Computing Architecture, Advanced Systems Lab, Algorithms Lab

ETH Zurich Sept. 2022 – Sept. 2025
BSc Computer Science - GPA: 5.15/6 *Zurich, Switzerland*

- **Thesis (6/6), Secure and Trustworthy Systems Group:** extended XRAY, a hardware security analysis tool; an automated pipeline ran **300,000+ test cases** and uncovered previously unknown vulnerabilities.
- **Relevant Coursework:** Algorithms & Data Structures/Probability, Information Retrieval, Introduction to Machine Learning, Data Modeling and Databases, Parallel Programming, Compiler Design, Operating Systems

Experience

ETH Zurich Feb – Jun 2024, 2025, 2026
Teaching Assistant - Digital Design and Computer Architecture *Zurich, Switzerland*

- Guided students through **10 hands-on lab sessions** on digital design, Verilog, MIPS assembly and FPGA.
- Reviewed submissions and **explained technical concepts** clearly to students with different backgrounds.

Projects

Real or AI? | *TypeScript, Astro, Svelte, Hono, Cloudflare Workers, D1, R2, Workers AI* | realoraiquiz.com

- **Launched a serverless real-vs-AI image game** on Cloudflare's edge (daily challenge, Blitz, leaderboards, accounts) at **$0 hosting cost**, by staying on free tiers and edge-caching every image for a year.
- **Built a nightly AI content pipeline** (GitHub Actions) that adds new photo/AI pairs across **~20 topics** within **~6,000 of 10,000 free daily AI neurons**: a vision model captions each public-domain photo and FLUX generates its twin.
- Server-side scoring, rate limits, hashed sign-in codes, CSP, **Llama Guard moderation**; **120+ tests** gate deploys.

FastGATr: High-Performance Transformer Kernels (team of 4) | *C++, ARM NEON SIMD, Python*

- **Achieved a 1.7-4.5x speedup over the optimised PyTorch reference** (**1500x over naive C++**) by generating sparse kernels (~140x fewer ops) and hand-vectorising with NEON intrinsics; explained the results with a **roofline model**.

Cloud Computing Architecture: Workload Scheduling (team of 3) | *Kubernetes, GCP, Python, OpenEvolve*

- **Met the p95 latency target in every run** while co-locating batch jobs with a latency-critical service on **Kubernetes (GCP)**, by measuring interference and writing a **Python controller** that adjusts CPU quotas live.
- **Evolved a scheduling policy 19% faster** than the hand-written one with **OpenEvolve**, comparing three LLMs; **prompt engineering** (explicitly allowing parallelism) was what got them past the starting solution.

Compiler for an Object-Oriented Language (course project) | *OCaml, LLVM IR, x86-64*

- Built a compiler for Oat: **type checking, LLVM IR to x86-64, dataflow optimisations, register allocation**.

KnowHouse | *Flutter, Dart, Firebase, GitHub Actions* | getknowhouse.com | Google Play

- **Built and shipped a family task app on Google Play**: task assignment, real-time sync, leaderboard, shared calendar.
- **200+ unit, provider and widget tests**; a one-click release workflow signs and uploads to Google Play.

Skills

Programming Languages: C++, Java, Python, C, TypeScript, JavaScript, SQL, Dart, OCaml, HTML, CSS
Cloud & Data: Google Cloud, Kubernetes, Docker, Cloudflare Workers, Firebase, PostgreSQL, GitHub Actions
AI/ML: LLM APIs, Workers AI, OpenEvolve, PyTorch
Frameworks & Tools: React, Svelte, Flutter
Languages: German (native), English (fluent), French (basic)